AYRA ERP SYSTEM REQUIREMENTS SPECIFICATION

Project Title: AYRA ERP Portal
Document Type: Detailed System Requirements Document

1. INTRODUCTION

AYRA ERP is a web-based enterprise resource planning platform developed for academic institutio
It supports daily university and college operations through separate role-based dashboards for students, teachers, academic office staff, and communication teams. The project centralizes academic data, student management, scheduling, communication, and alert broadcasting in a sing
integrated system.

2. PROJECT PURPOSE

The main purpose of this project is to reduce manual academic administration work and provide a central digital system where all major academic and operational workflows can be managed from o
place. The system improves visibility, data accuracy, access control, and communication speed.

3. PROJECT OBJECTIVES

- To provide role-based access for different categories of institutional users.
- To manage student records, subject lists, timetables, progress, and advisory communication.
- To help teachers and academic office staff coordinate academic operations efficiently.
- To provide structured leave request and approval workflows.
- To support alert-based communication using email delivery through Nodemailer.
- To maintain all major ERP records in MongoDB for persistence and retrieval.

4. SYSTEM SCOPE

The AYRA ERP system covers the following scope areas:

- Student self-service dashboard access
- Teacher academic operations and student management
- Academic office administration and oversight
- Communication management for notices, campaigns, and events
- Calendar and timetable visibility
- Leave request workflow
- Email alert creation, targeting, delivery, and history tracking

5. USER ROLES

5.1 Student

- View own profile details
- View subject list and class timings
- View teacher advisories and alerts
- Track semester progress, attendance, marks, and remarks
- View academic calendar events
- Apply for leave and track approval or rejection status
- Upload and remove profile image in dashboard profile area

5.2 Teacher

- Add, edit, and delete student records
- Maintain student emails for communication use
- Create and manage subject lists
- Create and manage class schedules and timings
- Publish advisories for students
- Update semester progress, marks, attendance, and remarks
- Approve or reject leave requests
- Send alerts by email to all students or selected groups
- Review previously sent alert records from the dashboard

5.3 Academic Office
- Add, edit, and delete teacher accounts
- Generate teacher passwords
- Maintain student registry visibility
- Create and manage curriculum plans
- Manage timetables and scheduling records
- Manage academic approvals and records
- Monitor leave requests
- Send and review alert messages to students

5.4 Communication Team
- Create and manage announcements
- Maintain communication campaigns
- Manage event records
- Track response data, escalation counts, and notes

6. MAJOR FUNCTIONAL REQUIREMENTS
6.1 Authentication and Session Management
- The system shall support login for multiple roles.
- The system shall use tenant-based route structure.
- The system shall preserve authenticated session details in frontend context.

6.2 Student Information Management
- The system shall store student ID, name, department, semester, section, email, phone, and statu
- The system shall allow teacher-side CRUD operations for student data.
- The system shall allow academic office review and update access for student registry data.

6.3 Academic Resources Management
- The system shall manage subject records with code, name, department, semester, credits, and fa
- The system shall manage class schedules with timings, room, section, and subject mapping.
- The system shall allow timetable visibility and calendar coordination.

6.4 Progress and Advisory Management
- The system shall record student attendance, marks, remarks, and advisor flags.
- The system shall display advisories with title, message, teacher name, and audience.

6.5 Leave Workflow
- Students shall be able to apply for leave using date and reason inputs.
- Teachers shall be able to accept or reject leave requests.
- Reject reason shall be stored and displayed when rejection occurs.
- Academic office shall be able to monitor leave records in read-only mode.

6.6 Alert and Email Communication
- Teachers and Academic Office shall be able to create alert messages.
- Alerts shall support types such as Exam, Event, Information, and Urgent.
- Alerts shall support all-students, department, semester, section, and specific-email targeting.
- Specific-email targeting shall validate against approved student records stored in MongoDB.
- Alert recipient count, recipient list, delivery status, and sent time shall be stored in database.
- The system shall send alert emails using Gmail credentials configured through environment varia

7. DATABASE REQUIREMENTS
The system uses MongoDB as the primary database. Major collections include:
- Users
- Students
- Teacher Subjects
- Class Schedules
- Advisories
- Student Progress
- Leave Requests
- Curriculum Plans
- Timetables
- Academic Approvals
- Academic Records
- Announcements
- Campaigns
- Calendar Events
- Response Tracking
- Teacher and Academic Alerts

Important stored alert fields include:
- tenantSlug
- alertType
- title
- message
- teacherName or academic sender name
- audienceType
- audienceValue
- targetAudience
- recipientEmails
- recipientCount
- deliveryStatus
- sentAt

- lastError

8. FRONTEND TECHNOLOGY REQUIREMENTS
- React 18.2.0
- React DOM 18.2.0
- React Router DOM 6.22.0
- Material UI 5.15.0
- Emotion React 11.11.0
- Emotion Styled 11.11.0
- Vite 5.0.0
- Tailwind CSS 3.4.0
- PostCSS 8.4.0
- Autoprefixer 10.4.0

9. BACKEND TECHNOLOGY REQUIREMENTS
- Node.js runtime
- Express 4.21.2
- Mongoose 8.10.0
- CORS 2.8.5
- Dotenv 16.4.5
- Nodemailer 6.9.16
- Nodemon 3.1.9 for development

10. THEME, FONT, AND DESIGN SYSTEM DETAILS
10.1 Typography
- Primary font family: Manrope, Segoe UI, sans-serif
- Typography h1 weight: 800
- Typography h1 letter spacing: -0.04em
- Typography h2 weight: 800
- Typography h2 letter spacing: -0.03em
- Typography h3 weight: 700
- Button font weight: 700
- Button text transform: none

10.2 Core Theme Colors
- Primary Main: #0f766e
- Primary Light: #ccfbf1
- Primary Dark: #115e59
- Secondary Main: #1d4ed8
- Background Default: #f4f7f4
- Background Paper: #ffffff
- Text Primary: #102a26
- Text Secondary: #47635d

10.3 Global Visual Styling
- Root font family: Manrope, Segoe UI, sans-serif

- Root text color: #102a26
- Root background: #f4f7f4
- Body gradient background: linear-gradient(180deg, #f8fafc 0%, #eef6f3 100%)
- Radial background accent: rgba(20, 184, 166, 0.1)
- Text selection color overlay: rgba(15, 118, 110, 0.18)
- Theme border radius: 18px
- Outlined input border radius: 16px
- Button border radius: 999px

10.4 Brand Accent Styling
- Logo gradient uses teal-600, emerald-500, and cyan-400
- Sidebar uses dark slate background styling
- Tenant workspace header uses glassmorphism white card styling
- Alert dashboard banner uses gradient #082f49 to #0f766e to #d97706

11. NON-FUNCTIONAL REQUIREMENTS
11.1 Usability
- The interface shall be clean, readable, and role-specific.
- The dashboard shall support responsive use across desktop and mobile screens.

11.2 Performance
- CRUD operations shall respond efficiently for normal institutional usage.
- Dashboard filtering and section switching shall be fast and readable.

11.3 Reliability
- Critical ERP records shall persist in MongoDB.
- Alert history and communication metadata shall be stored for tracking.

11.4 Security
- Sensitive credentials shall be managed using environment variables.
- Role-based access shall restrict dashboard features by user category.

11.5 Maintainability
- The codebase shall remain modular across frontend pages, components, routes, controllers, and

12. SOFTWARE REQUIREMENTS
- Operating System: Windows 10/11, Linux, or macOS
- Browser: Chrome, Edge, Firefox
- Package Manager: npm
- IDE: VS Code recommended

13. HARDWARE REQUIREMENTS
- Processor: Intel Core i3 / AMD equivalent or higher
- RAM: 4 GB minimum, 8 GB recommended
- Storage: At least 2 GB free disk space
- Internet: Required for mail delivery and cloud deployment

14. DEPLOYMENT REQUIREMENTS
- Frontend may be deployed on Vercel
- Backend may be deployed on Render or Railway
- Production deployment requires cloud MongoDB such as MongoDB Atlas
- Localhost MongoDB is suitable only for local development on the same machine
- Required environment variables include MONGODB_URI, DEFAULT_TENANT, MAIL_ALERT_E
  MAIL_ALERT_APP_PASSWORD, MAIL_ALERT_FROM_NAME, and frontend API base URL

15. CONCLUSION
AYRA ERP is a modular, database-driven university ERP platform that integrates academic workfl
communication management, user dashboards, and alert messaging into one centralized applicati
The system is suitable for institutions that need a lightweight but extendable ERP solution with
modern frontend design, MongoDB persistence, and role-based operations.